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Computational Methods of Optimization

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1 Introduction

1.1 References

- Numerical Optimization by J.Nocedal, S.Wright
- Practical Methods in Optimization by R.Fletcher

1.2 Some Optimization Examples

- **Univariate Optimizations**: Minimize $f(x)$ where

$$f(x) = ax^2 + bx + c,\ x \in \mathbb{R}$$

- Systematic procedure to find the minimum and maximum values of a function f on an interval $[a, b]$
  1. Solve $f'(x) = 0$ to find list of critical points of f
  2. Exclude any critical points not inside the interval $[a, b]$
  3. Add to the list the endpoints a, b of the interval and any points of discontinuity or non differentiability.
  4. At each of these points in the list, evaluate the function f: the biggest number that occurs is the maximum, and the smallest number that occurs is the minimum

$$f'(x) = 2ax + b = 0 \implies x^* = \frac{-b}{2a}$$

Minimum is that value for which

$$\begin{aligned} f(x) &\geq f(x^*),\ \forall x \\ \implies f(x^*) &= c - \frac{b^2}{4a} \end{aligned}$$

- For $u, v \in \mathbb{R}^d$, minimize $f(t)$

$$\begin{aligned} f(t) &= \frac{1}{2}\|u - tv\|^2 = \frac{1}{2}(u - tv)^T(u - tv) = \frac{1}{2}(u^T - tv^T)(u - tv) \\ &= \frac{1}{2}(u^Tu - (u^Tv)t - (v^Tu)t + t^2v^Tv) \\ &= \frac{1}{2}u^Tu - t(v^Tu) + \frac{t^2}{2}v^Tv \end{aligned}$$

Comparing with the previous problem we have

$$\begin{aligned} &a = \frac{v^Tv}{2}\ b = -v^Tu\ c = \frac{u^Tu}{2} \\ &t^* = \frac{-b}{2a} = \frac{v^Tu}{v^Tv} \qquad f(t^*) = \frac{u^Tu}{2} - \frac{(v^Tu)^2}{2v^Tv} \geq 0 \text{ as its representing distance} \\ &\implies \frac{(u^Tu)(v^Tv) - (v^Tu)^2}{2v^Tv} \geq 0 \implies (u^Tu)(v^Tv) \geq (v^Tu)^2 \\ &\implies \|u\|^2\|v\|^2 \geq (v^Tu)^2 \\ &\implies |v^Tu| \leq \|u\|\|v\| \leftarrow \text{ Cauchy Schwarz Inequality} \end{aligned}$$

This function will be zero when $f(t^*) = 0 \implies u = t^*v$.

- Having bounded, and closed sets provides nice minimums else we have issues of unboundedness and non-existence of minimum.

1.3 Some Useful Tools

- **Distance function**: $d(\cdot,\cdot) : X \times X \to \mathbb{R}$

  1. $d(x,z) > 0$ where $x \neq z$ and $d(x,x) = 0$
  2. $d(x,z) = d(z,x)$
  3. $d(x,z) \leq d(x,y) + d(y,z)$
  4. (X,d) is a metric space

  We will be using $d(x,y) = \|x-y\|$, Euclidean distance.

- **Neighborhood**: $N_r(x) = \{z | \|x-z\| < r\}$, where $x \in \mathbb{R}^d$. Circle of radius $'r'$ not considering the points on the circle.

- **Interior points**: A point $x \in A$ is an interior point of A if there exists $r > 0$ such that $N_r(x) \subset A$
  $Int(A) =$ Set of interior points of A

- **Open Set**: A set A is said to be an open set if $Int(A) = A$, every point is an interior point of A.

$$(a,b) = \{x | a < x < b\}$$

- **Limit Point**: A point x is called limit point of A if for every $r > 0$, $N_r(x) \cap A \neq \phi$. $Int(A)$ are all limit points.

- **Closed Set**: A set A is said to be closed set, if all limit points of A are in A.

- **Closure of a Set**: $Closure(C) = C \cup \{\text{limit points of } C\}$

- **Bounded Sets**: If there exists $M > 0$ and $x \in \mathbb{R}^d$ such that $\|x-z\| < M$ for all $z \in C$ then the set C is bounded.

- Let v be an upper bound and if any $\hat{v} < v$ is not an upper bound, then v is called the least upper bound of A. $v = sup\ A$.

- If A is bounded below, w is a lower bound and any $\hat{w} > w$ is not a lower bound then w is called the greatest lower bound. $w = inf\ A$.

- **Limit Point**: x is a limit point of C if $\forall \delta > 0$, $N_\delta \cap C \neq \phi$.

- If x_0 is a limit point of C, then $\exists$ a sequence $\{x^n\}$ such that

$$\lim_{n\to\infty} x^{(n)} = x_0$$

- **Limits**: $f : C \subseteq \mathbb{R}^d \to \mathbb{R}$

$$\lim_{x\to z} f(x) = p$$

  For every $\epsilon > 0$, $\exists \delta > 0$, such that $\forall x \in N_\delta(z)$

$$|f(x) - p| < \epsilon$$

- Let $f(x)$ and $g(x)$ be functions $\mathbb{R}^d \to \mathbb{R}$ where $g(x) \neq 0 \forall x$

$$\lim_{x\to x_0} \frac{f(x)}{g(x)} = 0$$

  then $f(x) \in o(g(x))$ at $x = x_0$

- $\forall \epsilon > 0$, $\exists \delta < 0$ such that $\forall \|x - x_0\| < \delta$ and $|\frac{f(x)}{g(x)}|\epsilon$

- **Continuous function**: $f : C \subseteq \mathbb{R}^d \to \mathbb{R}$ is continuous at $x = x_0$ if

$$\lim_{x\to x_0} f(x) = f(x_0)$$

  $\forall \epsilon$, $\exists \delta$ such that

$$|f(x) - f(x_0)| < \epsilon,\ \forall \|x - x_0\| < \delta$$

  f is continuous on C if f is continuous at every $x \in C$.

- Let f be a continuous at x_0 $h(x) = f(x) - f(x_0)$

$$\lim_{x \to x_0} h(x) = 0,\ h(x) \in o(1),\ f(x) = f(x_0) + o(1)$$

- **Intermediate Value Theorem**: If f is continuous on $[a, b]$, $f : \mathbb{R} \to \mathbb{R}$ and suppose $f(a) < f(b)$ then $\forall f(a) < h < f(b)$ there exists $x \in (a, b)$ such that $f(x) = h$

- **Weierstrass**: Let C be closed and bounded set if $f : C \to \mathbb{R}$ be continuous, then $\exists p, q \in C$ such that $f(p) = m = inf\ f(C)$, $f(q) = M = sup\ f(C)$

$$f(p) \leq f(x) \leq f(q),\ \forall x \in C$$

  This guarantees the existence of p and q for which min and max can be obtained.

- **Strategy: Iterative Scheme**: Consider the sequence $x^{(i)}$

  1. Is $x^{(i)}$ the desired point: stop if it is and if not iterate
  2. How will we know $x^{(i)}$ is the desired point: it's hard
  3. Assume we have a test given a point it will tell it is desired or not.
  4. If given point is not a desired point: Build an approx to $f(x)$ around $x^{(i)}$, choose $x^{(i+1)}$.

- **Derivative**: $\frac{f(x)-f(x_0)}{x-x_0} = m(x; x_0)$

$$f(x) = f(x_0) + m(x; x_0)(x - x_0)$$

  Consider we know "m", $\lim_{x \to x_0} m(x; x_0) = m(x_0)$

$$f'(x) = \lim_{t \to x} \frac{f(t) - f(x)}{t - x} = \lim_{t \to x} g(t)$$

  If $f'(x)$ exists then f is said to be differentiable at x.

- If x^* is a local minimum, then $f'(x^*) = 0$ if $f'(x^*)$ exists.

- Let $f : [a, b] \to \mathbb{R}$ be differentiable in (a, b). There is a point $z \in (a, b)$ such that

$$f(b) - f(a) = (b - a)f'(z)$$
$$h(x) = (f(b) - f(a))x - (b - a)f(x)$$

- **Taylor Series**: $f^{(i+1)}(x) = \frac{d}{dx} f^{(i)}(x)$

$$f(y) = P_{n-1}(y; x) + R_n(y) = f(x) + \sum_{i=1}^{n-1} f^{(i)}(x) \frac{(y - x)^i}{i!}$$

$$\text{For } n = 1$$
$$f(y) = f(x) + f'(z)(y - x)$$
$$\text{For } n = 2$$
$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}(y - x)^2 f''(z)$$

- $C^{(n)}$: Set of all function with $f^{(n)}$ being continuous.

- Consider $f \in C^{(1)}$, f is a function with f' being continuous and 2nd derivative may or may not exist.

$$f(y) = f(x) + f'(x)(y - x) + o(1)(y - x) = P_1(y) + o(y - x)$$

- Now consider $f \in C^{(2)}$, f'' is continuous

$$f(y) = P_2(y) + o((y - x)^2) = f(x) + (y - x)f'(x) + \frac{(y - x)^2}{2!} f''(x) + o((y - x)^2)$$

- **Hessian**: 2nd derivative of all the partial derivatives, $H(x) \in \mathbb{R}^{d \times d}$ is a $d \times d$ matrix

- If $f \in C^{(2)}$, $H(x)$ is a symmetric matrix.

- **Directional Directive**: Directional derivative of x along u, $\nabla f(x)^T u$.

2 Unconstrained Optimization

2.1 Introduction

- Consider $f \in C^{(1)}$ $\exists \delta > 0$ such that

$$f(y) = f(x) + \nabla f(x)^T (y - x), \ \forall y \in N_\delta(x)$$

Since x^* is the local minima, then we have $f(x) \geq f(x^*)$. From Taylor series we have

$$\begin{aligned} P_1(x; x^*) &\geq f(x^*) \\ f(x^*) + \nabla f(x^*)^T (x - x^*) &\geq f(x^*) \\ \nabla f(x^*)^T (x - x^*) \geq 0 &\implies \nabla f(x^*) = 0 \end{aligned}$$

- Consider $f \in C^{(2)}$, let x^* be a local minimum $f \in C^{(2)} implies f \in C^{(1)} \implies \nabla f(x^*) = 0$, as before

$$f(x^*) + \frac{1}{2}(x - x^*)^T H(x^*)(x - x^*) \geq f(x^*) \implies u^T H(x^*) u \geq 0$$

$H(x^*)$ is a positive semi definite, symmetric and real valued, $H(x^*) \in S_d$

- **Sufficient Condition**: $f \in C^{(2)}$ and x^* is a point, if $\nabla f(x^*) = 0$ and $H(x^*) > 0$ then x^* is a local minimum.
Any point x^* which satisfies these two conditions is a local minimum.

- **Coercive function**: Function is continuous, if x is either large in positive or negative direction then $f(x)$ is also very large

$$\lim_{\|x\| \to \infty} f(x) = \infty, \ f : \mathbb{R}^d \to \mathbb{R}$$

Magnitude(Norm) of x is large then $f(x)$ is large.

- If f is coercive and $C^{(1)}$ then the set of global minimum is non-empty and is attained at some of the critical points.

- **Convex Set**: $S \subseteq \mathbb{R}^d$ is said to be a convex set, if for every $x^{(1)}, x^{(2)} \in S$ and for every $0 < \theta < 1$

$$\theta x^{(1)} + (1 - \theta) x^{(2)} \in S$$

$x^{(1)}, x^{(2)}, ..., x^{(n)} \in S$

$$\lambda_1 x^{(1)} + \lambda_2 x^{(2)} + ... + \lambda_n x^{(n)} \in S, \ \sum_{i=1}^{n} \lambda_i = 1$$

Any x written like this is called as convex combination of $x^{(1)}$ through $x^{(n)}$.
Intersection of convex sets may be convex sets.

- Let S be a convex set, $f : S \subseteq \mathbb{R}^d \to \mathbb{R}$ is a convex function if for any $x^{(1)}, x^{(2)} \in S$

$$f((1 - \alpha) x^{(1)} + \alpha x^{(2)}) \leq (1 - \alpha) f(x^{(1)}) + \alpha f(x^{(2)}), \text{ holds } \forall 0 < \alpha < 1$$

- Any critical point is the global minimum for a convex function

2.2 Iterative Algorithm

- Consider the iteration scheme

  1. find $u^{(k)}$
  2. find $\alpha^{(k)}$
  3. Set $x^{(k+1)} = x^{(k)} + \alpha^{(k)} u^{(k)}$
  4. Check for local minimum

Here α is the step size, and u is the descent direction.
For u to be a descent direction, it must satisfy $\nabla f(x)^T u < 0$

- For a descent direction u at x we have

$$f(x+\alpha u) < f(x), \text{ for } 0 < \alpha < \delta$$

Form a function $g : \mathbb{R} \to \mathbb{R}$, $g(\alpha) = f(x + u\alpha)$, Since $f \in C^{(1)}$ and $g \in C^{(1)}$

$$\implies g(\alpha) = g(0) + \alpha g'(0)$$

We need an α such that $g(\alpha) < g(0)$

- One descent direction we can take is $u = -\nabla f(x)$. This is also called the **steepest descent direction**. We will get max value when $u = \nabla f(x)$ and min value when $u = -\nabla f(x)$
- Consider minimizing quadratic function $f(x) = \frac{1}{2}x^T Qx - b^T x$, where $Q \in S_d^+$.

$$\nabla f(x) = Qx - b \text{ and } H(x) = Q$$

Since, $H(x)$ is positive, the function is convex and critical point is global minima, attained at $x^* = Q^{-1}b$. To find this x^*, we need to find Q^{-1}, which is expensive.
To avoid inversion, we try to find an iterative algorithm.

2.3 Building Blocks of Iterative Algorithm

- If $a, b \in \mathbb{R}^d$, then from Cauchy Schwarz inequality $|a^T b| \leq \|a\|^2 \|b\|^2$.
- For any symmetric matrix A, for any $u \in \mathbb{R}^d$, there exists such that

$$u^T Au \leq L\|u\|^2$$

- Consider minimizing the function $f(x) = ax^2 + bx + c$

$$x^* = \frac{-b}{2a}, \; f(x^*) = c - \frac{b^2}{4a}$$

Eliminating b we get

$$f(x) = ax^2 - (2ax^*)x + c = a(x^2 - 2xx^*) + c$$

- Consider minimizing $h(x) = \frac{1}{2}(x + \frac{a}{x})$. We know from AM-GM inequality

$$\frac{1}{2}(a+b) \geq \sqrt{ab}$$
$$h(x) \geq \sqrt{x}(\frac{\sqrt{a}}{\sqrt{x}}) = \sqrt{a}$$

Minimum attained at $x = \sqrt{a}$, and $h(x) = \sqrt{a}$

- Consider $Q \in S_d^+$, let $\lambda_1 \geq \lambda_2 \geq ... \geq \lambda_d > 0$ are d eigen values.
Define $B = Q + cQ^{-1}$, $c > 0$. Eigen values of B are $\lambda_i(B) = \lambda_i + \frac{c}{\lambda_i}$

$$\lambda_i \leq \lambda_i(B) \leq \lambda_1 + \frac{c}{\lambda_d}$$

For any $u \in \mathbb{R}^d$

$$u^T Bu = u^T Qu + cu^T Q^{-1} u$$

By AM-GM Inequality

$$u^T Bu \geq 2\sqrt{(u^T Qu)(cu^T Q^{-1} u)}$$
$$(\lambda_1 + \frac{c}{\lambda_d})\|u\|^2 \geq u^T Bu \geq 2\sqrt{(u^T Qu)(cu^T Q^{-1} u)}$$
$$(\lambda_1 + \frac{c}{\lambda_d})\|u\|^2 \geq 2\sqrt{(u^T Qu)(cu^T Q^{-1} u)}$$
$$(\lambda_1 + \frac{c}{\lambda_d})^2\|u\|^4 \geq 4c(u^T Qu)(u^T Q^{-1} u)$$

Consider $c = \lambda_1 \lambda_d$

$$(\lambda_1 + \lambda_d)^2\|u\|^4 \geq 4\lambda_1\lambda_d(u^T Qu)(u^T Q^{-1} u)$$
$$\frac{\|u\|^4}{(u^T Qu)(u^T Q^{-1} u)} \geq \frac{4\lambda_1\lambda_d}{(\lambda_1 + \lambda_d)^2}$$

This result is **Kontrovich Inequality**

2.4 Iteration Schemes

- Revisiting our problem, consider $y = x + \alpha u$

$$
\begin{aligned}
&f(y) = f(x) + \alpha \nabla f(x)^T u + \frac{\alpha^2}{2} u^T Q u \\
&g(\alpha) = f(x + \alpha u) = f(y): \text{ convex function} \\
&f(y) \text{ is of the form } y = ax^2 + bx + c, \text{ min at } \frac{-b}{2a} \\
&\text{min is at } \alpha^* = \frac{-\nabla f(x)^T u}{u^T Q u}, \text{ Eliminating 'b' or } \nabla f(x)^T u = -\alpha^* u^T Q u \\
&f(y) = \alpha^2 (\frac{1}{2} u^T Q u) + \alpha(-\alpha^* u^T Q u) + f(x) \\
&f(x + \alpha u) = f(x) + (\alpha^2 - 2\alpha\alpha^*) \frac{u^T Q u}{2} \\
&f(x + \alpha^* u) = f(x) - \frac{(\alpha^*)^2}{2} u^T Q u \\
&f(x + \alpha^* u) \leq f(x + \alpha u) \\
&x \to x + \alpha^* u, \ f(x) \to f(x + \alpha^* u) \\
&f(x) \to f(x) - \frac{(\alpha^*)^2}{2} u^T Q u \implies \text{ decrease in function value is } \frac{(\alpha^*)^2}{2} u^T Q u
\end{aligned}
$$

$f(x)$ is bounded from below at $x^* = Q^{-1} b$, so we can't keep on decreasing it forever
If we repeat at every point the function value keeps on decreasing by $-\frac{(\alpha^*)^2}{2} u^T Q u$
At a given x for a fixed u, the best value of α for which $f(y)$ is min is given by

$$f(y) = f(x) + \alpha \nabla f(x)^T u + \frac{\alpha^2}{2} u^T Q u$$

- Now coming to the iteration scheme

$$
\begin{aligned}
&x^{(k+1)} = x^{(k)} - \alpha_k \nabla f(x^{(k)}) \\
&u = -\nabla f(x), \ \nabla f(x^{(k)}) = g_k, \text{ and } \alpha_k = \frac{g_k^T g_k}{g_k^T Q g_k} \\
&\implies f(x^{(k+1)}) = f(x^{(k)}) - \frac{1}{2} \frac{(g_k^T g_k)^2}{g_k^T Q g_k}
\end{aligned}
$$

Repeat this process many times

$$
\begin{aligned}
&f(x) = f(x^*) + \frac{1}{2}(x - x^*)^T Q (x - x^*) = f(x^*) + E(x) \\
&\nabla f(x) = Qx - b = Q(x - x^*) \\
&\nabla f(x^{(k)}) = Q(x^{(k)} - x^*) = g_k, \implies x^{(k)} - x^* = Q^{-1} g_k \\
&\text{now } E(x^{(k)}) = \frac{1}{2}(x^{(k)} - x^*)^T Q (x^{(k)} - x^*) = \frac{1}{2} g_k^T Q^{-1} g_k
\end{aligned}
$$

Now we have

$$
\begin{aligned}
&f(x^{(k)}) = f(x^*) + E(x^{(k)}) \\
&f(x^{(k+1)}) = f(x^*) + E(x^{(k+1)}) \\
&E(x^{(k+1)}) - E(x^{(k)}) = f(x^{(k+1)}) - f(x^{(k)}) = \frac{-1}{2} \frac{(g_k^T g_k)^2}{g_k^T Q g_k} \\
&\frac{E(x^{(k)}) - E(x^{(k+1)})}{E(x^{(k)})} = \frac{(g_k^T g_k)^2}{(g_k^T Q g_k)(g_k^T Q^{-1} g_k)}
\end{aligned}
$$

By Kontrovich Inequality

$$
\begin{aligned}
&\frac{E(x^{(k)}) - E(x^{(k+1)})}{E(x^{(k)})} \geq \frac{4\lambda_1 \lambda_d}{(\lambda_1 + \lambda_d)^2} \\
&E(x^{(k+1)}) \leq [1 - \frac{4\lambda_1 \lambda_d}{(\lambda_1 + \lambda_d)^2}] E(x^{(k)}) \\
&E(x^{(k+1)}) \leq (\frac{\lambda_1 - \lambda_d}{\lambda_1 + \lambda_d})^2 E(x^{(k)})
\end{aligned}
$$

Consider $\rho = \frac{\lambda_1 - \lambda_d}{\lambda_1 + \lambda_d}$

$$\begin{aligned} & E(x^{(k+1)}) \leq \rho^2 E(x^{(k)}) \\ \implies & E(x^{(k)}) \leq \rho^{2k} E(x_0) \\ & \lim_{k\to\infty} E(x^{(k)}) = 0, \ \lim_{k\to\infty} x^{(k)} = x^* \end{aligned}$$

2.5 Line Search

- We take $u^{(k)} = -\nabla f(x^{(k)})$, the steepest descent direction.
- Now, we are interested in finding the optimal step size α_k.
- **Exact Line Search**

$$\arg\min_{\alpha \geq 0} f(x^{(k)} + \alpha u^{(k)})$$

  Exact line search is resource intensive and can't be done in every iteration, so in general we will consider inexact line search

- **Inexact Line Search**

$$\begin{aligned} & g(\alpha) = f(x^{(k)} + \alpha u^{(k)}), \ x^{(k)} \text{ and } u^{(k)} \text{ are fixed} \\ & g(0) = f(x^{(k)}) \\ & g'(0) = \nabla f(x^{(k)})^T u^{(k)} : \text{ Directional derivative} \\ & u^{(k)} \text{ is a descent direction at } x^{(k)} \end{aligned}$$

  New point should have two properties

$$\begin{aligned} & \alpha > 0 \text{ should be large} \\ & f(x^{(k)} + \alpha u^{(k)}) < f(x^{(k)}) \text{ decrease in function value} \end{aligned}$$

  We have the Wolfe conditions

$$\begin{aligned} & g(\alpha) \leq g(0) + \alpha \rho_1 g'(0) \\ & g'(\alpha) \geq \rho g'(0) \\ & \rho \geq \rho_1 \end{aligned}$$

- **The Gradient Descent Algorithm** $f \in C^{(1)}$ and first order model. We first initialize $x^{(0)}$
  At $x^{(k)}$ find $u^{(k)}$ such that $\nabla f(x^{(k)})^T u^{(k)} < 0$
  Find step size through line search

$$x^{(k+1)} \leftarrow x^{(k)} + \alpha_k u^{(k)}$$

  Goldstein condition, $\rho_2 = 1 - \rho_1$

$$\{\alpha | g(0) + \alpha \rho_2 g'(0) \leq g(\alpha) \leq g(0) + \alpha \rho_1 g'(0)\}$$

2.6 Second order Preliminaries

- Let $s_n \in \mathbb{R}$ for $n \geq 1$, and $s_n \leq s_{n+1}$, montone increasing
  The sequence $\{s_n\}$ converges iff its bounded

$$\lim_{n\to\infty} s_n = s^*$$

- Let $\{t_i\}$ be a sequence such that $t_i \geq 0$ for all $i \geq 1$

$$s_n = \sum_{i=1}^{n} t_i, \text{ clearly } s_n \leq s_{n+1}$$

$$\lim_{n\to\infty} s_n = s^* \text{ then } \lim_{n\to\infty} t_n = 0$$

- Consider minimizing a function $f(x)$

$$\begin{aligned} & f(x) \geq M, \ \forall x, \ f \text{ is bounded from below} \\ & f \in C^{(1)} \text{ if } \exists L > 0 \\ & \|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \ \forall x, y \in \mathbb{R}^d \to \textbf{Lipschitz Condition} \\ & f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2}\|x - y\|^2 \end{aligned}$$

- **Global Convergence Theorem**: The gradient descent algorithm when applied to $f \in C^{(1)}$

$$\|\nabla f(x^{(k)})\| \to 0, \text{ under assumptions } \cos^2\theta^{(k)} > 0$$

- $A \in S_d$, we define spectral norm as

$$\|A\| = \max_i |\lambda_i|$$

  λ_i: Eigen values of A, and u_i are eigen vectors of A
  Eigenvectors are basis

$$\|Ax\| \leq \|A\|\|x\|$$

- Consider $f \in C_M^{(2)} \implies H(x) \in S_d$

$$\implies \|H(x) - H(y)\| \leq M\|x - y\|, \ \forall x, y \in \mathbb{R}^d$$

  Let $r(x) = \|x - y\|$, $B(x, y) = H(x) - H(y)$

$$\begin{aligned}
&\|B(x,y)\| = \max_j |\lambda_i(B(x,y))| \\
&|\lambda_i(B)| \leq Mr(x) \\
&-Mr(x) \leq \lambda_i(B) \leq Mr(x), \ \forall i \\
&-Mr(x)I \leq H(x) - H(y) \leq Mr(x)I \\
&H(y) - Mr(x)I \leq H(x) \leq H(y) + Mr(x)I
\end{aligned}$$

  $H(y)$ and $H(x)$ are both positive definite

$$\begin{aligned}
&H(y) \geq aI, \ a > 0 \\
&(a - Mr(x))I \leq H(x) \\
&a - Mr(x) \geq 0 \implies r(x) = \|x - y\| \leq \frac{a}{M}
\end{aligned}$$

- For $C_M^{(2)}$ function $H(y)$ is positive definite is being carried to its neighboring points by smoothness and this happens in the neighborhood of length "$\frac{a}{M}$"
- If $g(t) = f(y + t(x - y))$, and $g'(t) = \nabla f(y + t(x - y))^T(x - y)$

$$\begin{aligned}
&\int_0^1 g'(t)dt = g(1) - g(0) \\
&f(x) - f(y) = \int_0^1 \nabla f(y + t(x - y))^T(x - y)dt
\end{aligned}$$

  Similarly,

$$\nabla f(x) - \nabla f(y) = \int_0^1 H(y + t(x - y))(x - y)dt$$

2.7 Newton Method

- Assume $f \in C_M^2$, $f(x) = f(y) + \nabla f(y)^T(x - y) + \frac{1}{2}(x - y)^T H(y)(x - y) + o(\|x - y\|^2)$
- So, our goal is minimize $\frac{1}{2}u^T Qu + u^T b + c$,
  we know the solution to this is $u = -Q^{-1}b$ or $x = y - (H(y))^{-1}\nabla f(y)$

$$f(y - (H(y))^{-1}\nabla f(y)) \leq f(x)$$

- Newton iterates: $x^{(k+1)} = x^{(k)} - (H(y))^{-1}\nabla f(y)$

$$\begin{aligned}
&\nabla f(x^{(k)})^T u^{(k)} < 0 \\
&\nabla f(x^{(k)})^T (H(y))^{-1} \nabla f(x^{(0)}) < 0
\end{aligned}$$

  So, as long as $H(x^{(k)}) > 0$, $u^{(k)}$ is a descent direction.

- **Analysis**: Assume $f \in C_M^2$, $H(x^*) > aI$, and $\nabla f(x^*) = 0$. We can write

$$x^{(k+1)} - x^* = x^{(k)} - x^* - (H(x^{(k)}))^{-1}\nabla f(x^{(k)})$$

From earlier we know

$$\nabla f(x^*) - \nabla f(x^{(k)}) = \int_0^1 H(x^{(k)} + t(x^* - x^{(k)}))(x^* - x^{(k)})dt$$

Now, $\nabla f(x^*) = 0$, then we can substitute the rest

$$x^{(k+1)} - x^* = (x^{(k)} - x^*) + (H(x^{(k)}))^{-1}[\int_0^1 H(x^{(k)} + t(x^* - x^{(k)}))(x^* - x^{(k)})dt]$$

We can rearrange to get

$$x^{(k+1)} - x^* = (H(x^{(k)}))^{-1}[H(x^{(k)})(x^{(k)} - x^*) + \int_0^1 H(x^{(k)} + t(x^* - x^{(k)}))(x^* - x^{(k)})dt]$$

Take $A(t, x, x^*) = H(x^{(k)} + t(x^* - x^{(k)})) - H(x^{(k)})$, then using Cauchy Schwartz we get

$$\|x^{(k+1)} - x^*\| \leq \|(H(x^{(k)}))^{-1}\| \int_0^1 \|A(t, x^{(k)}, x^*)\|\|x^* - x^{(k)}\|dt$$

- If $H(x^{(k)}) \geq lI$ then $(H(x^{(k)}))^{-1} \leq \frac{1}{l}I$
  Since, $(H(x^{(k)}))^{-1} > 0$, $\implies \|H(x^{(k)})^{-1}\| \leq \frac{1}{l}$
- Using the above and the fact that $f \in C_M^2$ we have

$$\|x^{(k+1)} - x^*\| \leq \frac{M}{l}\int_0^1 t\|x^* - x^{(k)}\|^2 dt = \frac{M}{2l}\|x^* - x^{(k)}\|^2$$

Let $r_k = \|x^{(k)} - x^*\|$, we know that $H(x^{(k)}) \geq (a - Mr_k)I$ where $a - Mr_k$ is the value of l

$$\implies r_{k+1} \leq \frac{M}{2(a - Mr_k)}r_k^2 \leftarrow \text{ Quadratic Convergence}$$

- Now, $\frac{Mr_k}{2(a-Mr_k)} < 1$ for $r_{k+1} < r_k$

$$Mr_k < 2a - 2Mr_k \implies r_k < \frac{2}{3}\frac{a}{M} \implies r_{k+1} < \frac{2}{3}\frac{a}{M}$$

So, let $r_0 < \frac{2}{3}\frac{a}{M}$, $x^{(0)}$ is not arbitrary close to x^*. In such a case it is guaranteed that $r_{k+1} < r_k$ $\forall k \geq 0$.

- $\frac{2}{3}\frac{a}{M}$ is called newton region, if we choose a point in the newton region there will be quick convergence.
- If we are far from Newton region, it's better to use gradient method.
- a is the smallest eigen value of $H(x^*)$ but we don't know x^*. So it's better to start with gradient and if the values are not changing after few iterations then use newton's method.
- If $r_0 < \frac{2}{3}\frac{a}{M}$ then $r_k < r_0$, $H(x^{(k)}) > 0$ and

$$\|x^{(k+1)} - x^*\| \leq \frac{M\|x^{(k)} - x^*\|^2}{2(a - M\|x^{(k)} - x^*\|)}$$

2.8 Conjugate Gradient Descent

- Consider $f(x) = \frac{1}{2}x^TQx + b^Tx$, $Q \in S_Q^+$, $x \in \mathbb{R}^d$
- Consider a set of linearly independent directions $S = \{u^1, u^2, ..., u^d\}$ such that

$$(u^{(i)})^TQu^{(j)} = 0,\ i \neq j,\ Q \in S_Q^+$$

We call such a set of vectors Q-conjugate

- For $f(x) = \frac{1}{2}x^TQx + b^Tx$, $g_k = \nabla f(x^{(k)}) = Qx^{(k)} + b$.
  We know from exact line search that

$$\arg\min_{\beta} f(x^{(k)} + \beta(u^{(k)})) \implies \beta_k = -\frac{g_k^T u^{(k)}}{(u^{(k)})^T Q u^{(k)}}$$

  We can have an iteration scheme as

$$\begin{aligned}
&x^{(k+1)} = x^{(k)} + \beta_k u^{(k)} \leftarrow \text{ Conjugate Direction Iteration(CDI)} \\
&x^{(k)} = x^{(0)} + \sum_{i=0}^{k-1} \beta_i u^{(i)} \\
&Q(x^{(k)} - x^{(0)}) = \sum_{i=0}^{k-1} \beta_i Q u^{(i)} \\
&u^{(k)T} Q(x^{(k)} - x^{(0)}) = 0
\end{aligned}$$

- Since $u^{(i)}$s are L.I, they form a basis for $\mathbb{R}^d$. So,

$$x^* - x^{(0)} = \sum_{i=0}^{d-1} \alpha_i u^{(i)}$$

- If we prove that $\alpha_i = \beta_i$, then we can say that in the CDI, $x^{(d)} = x^*$

$$x^* - x^{(k)} + x^{(k)} - x^P(0) = \sum_{i=0}^{d-1} \alpha_i u^{(i)}$$

  Multiply by Q and $u^{(k)T}$

$$(u^{(k)})^T Q(x^* - x^{(k)}) + (u^{(k)})^T Q(x^{(k)} - x^{(0)}) = \alpha_k u^{(k)T} Q u^{(k)}$$

  Now, $g_k = \nabla f(x^{(k)} = Q(x^{(k)} + b = Q(x^{(k)} - x^*)$

$$\begin{aligned}
-u^{(k)T} g_k &= \alpha_k u^{(k)T} Q u^{(k)} \\
\alpha_k &= \beta_k
\end{aligned}$$

  Thus, $x^{(d)} = x^*$ it takes exactly d iterations to reach minima

- If $Q = I$, then standard basis can be used as Q conjugate direction.
  β_i minimizes i^{th} dimension or i^{th} candidate. This is called **coordinate wise minimization**
- After "d" iterations we have minimized in all "d" dimensions and got x^*
- How to find such a set of $u^{(i)}$s ?

2.9 Expanding Subspace Theorem

- Let $x^{(k+1)} = x^{(k)} + \beta_k u^{(k)}$ as in CDI. Then $g_k^T u^{(i)} = 0$, $x^{(k)} = \arg\min_{x \in B_k} f(x)$ where

$$B_k = \{x^{(0)} + \sum_{i=0}^{k-1} r_i u^{(i)} | \forall i \in \mathbb{R}\}$$

- Let $\min_{x \in L} f(x)$, where $L = \{x = x^{(k)} + \alpha u^{(k)}\}$

$$\begin{aligned}
f(x) &= f(x^{(k)}) + \nabla f(x^{(k)})^T (x - x^{(k)}) + \frac{1}{2}(x - x^{(k)})^T Q (x - x^{(k)}) \\
&= f(x^{(k)}) + \alpha(Qx^{(k)} + b)^T u_k + \frac{1}{2}\alpha^2 u_k^T Q u_k = h(\alpha)
\end{aligned}$$

  We have $\min_{x \in L} f(x) = \min_{\alpha \in \mathbb{R}} h(\alpha)$; $x^* = x^{(k)} + \alpha^* u^{(k)}$
  Now, $h''(\alpha) = u_k^T Q u_k > 0$ then $h'(\alpha) = 0 \implies \alpha^*$ is global minima

$$\begin{aligned}
h'(\alpha^*) &= g_k^T u_k + \alpha^* u_k^T Q u_k = 0 \\
\alpha^* &= -\frac{g_k^T u_k}{u_k^T Q u_k} = \beta_k
\end{aligned}$$

- $x \in B_{k+1} = \{x^{(0)} + \sum_{i=0}^{k} r_i u^{(i)}\}$; $\min_{x \in B_{k+1}} f(x)$. Again do substitution

$$f(x) = f(x^{(0)}) + \nabla f(x^{(0)})^T (x - x^{(0)}) + \frac{1}{2}(x - x^{(0)})^T Q (x - x^{(0)})$$

For any $x \in B_{k+1}$

$$f(x) = f(x^{(0)}) + g_0^T (\sum_{i=0}^{k} r_i u^{(i)}) + \frac{1}{2}(\sum_{i=0}^{k} r_i u^{(i)})^T Q (\sum_{i=0}^{k} r_i u^{(i)})$$

$$\Phi(r_0, ..., r_k) = f(x^{(0)}) + \sum_{i=0}^{k} r_i (g_0^T u^{(i)}) + \frac{1}{2} \sum_{i=0}^{k} r_i^2 u^{(i)T} Q u^{(i)}$$

Hessian of Φ is diagonal with positive values as $u^{(i)T} Q u^{(i)} \geq 0$. Now,

$$\frac{\partial \Phi}{\partial r_i} = 0$$
$$g_0^T u_i + (u_i^T Q u_i) r_i^* = 0$$
$$r_i^* = -\frac{g_0^T u_i}{u_i^T Q u_i}$$

Now, $g_i - g_0 = Q(x^{(i)} - x^{(0)}) = Q \sum_{l=0}^{i-1} r_l^* u_l$

$$u_i^T (g_i - g_0) = u_i^T Q (x^{(i)} - x^{(0)}) = \sum_{l=0}^{i-1} r_l^* u_i^T Q u_l = 0$$
$$u_i^T g_i = u_i^T g_0 \implies r_i^* = \beta_i$$

Therefore, we have $\Phi(r_0, ..., r_k) \geq \Phi(r_0^*, ..., r_k^*)$

$$x^{(k+1)} = x^{(0)} + \sum_{i=0}^{k} r_i^* u^{(i)}, \text{ and } f(x^{(k+1)}) \leq f(x) \ \forall x \in B_{k+1}$$

- **Theorem**: $B_k \subset B_{k+1}$, and $B_d = \mathbb{R}_d$
- **Conjugate Gradient Algorithm**: This is a special case of CDI in which we can calculate $u^{(k)}$ by using $u^{(k-1)}$
  Let $u_0 = -g_0$

$$x^{(k+1)} = x^{(k)} + \beta_k u_k, \text{ and } u_{k+1} = -g_{k+1} + r_k u_k$$

  where

$$\beta_k = -\frac{u_k^T g_k}{u_k^T Q u_k}, \text{ and } r_k = \frac{g_{k+1}^T Q u_k}{u_k^T Q u_k}$$

- Conjugate Gradient Algorithm is Conjugate Descent Algorithm.

2.10 Matrix Polynomial

- A polynomial is written as $P_k(z) = a_0 + a_1 z + a_2 z^2 + ... a_k z^k$
- Matrix polynomial of a square matrix Q is written as

$$P_k(Q) = a_0 I + a_1 Q + a_2 Q^2 + ... + a_k Q^k$$

- Now, we have $x^{(k+1)} = x^{(k)} + \alpha_k u_k = x^{(0)} + \sum_{i=0}^{k} \alpha_i u_i$
- $f(x) = \frac{1}{2} x^T Q x - b^T x = f(x^*) + E(x)$, where $E(x) = \frac{1}{2}(x - x^*)^T Q (x - x^*)$
- Recall that $x^{(k+1)} = \arg\min_{x \in B_{k+1} f(x)}$

$$x \in B_{k+1} \implies x = x^{(0)} + \sum_{i=1}^{k} \tilde{\alpha}_i u_i$$
$$= x^{(0)} + \sum_{i=0}^{k} \eta_i Q^i g_0$$
$$\implies x - x^* = x^{(0)} - x^* + (\sum_{i=0}^{k} (\eta_i Q^i g_0))$$

- Now, $g_0 = Q(x^{(0)} - x^*)$

$$x - x^* = (x^{(0)} - x^*) + \sum(\eta_i Q^i)Q(x^{(0)} - x^*)$$
$$= [I + (\sum_{i=0}^{k} \eta_i Q^i)Q](x^{(0)} - x^*)$$
$$x - x^* = [I + P_k(Q)Q](x^{(0)} - x^*)$$

- Now, $E(x) = \frac{1}{2}(x^{(0)} - x^*)^T[I + P_k(Q)Q]^T Q[I + P_k(Q)Q](x^{(0)} - x^*)$

$$\min_{x \in B_{k+1}} E(x) = \min_{P_k(Q)} \frac{1}{2}(x^{(0)} - x^*)^T Q[I + P_k(Q)Q]^2(x^{(0)} - x^*)$$

- Now, we need to minimize over a matrix polynomial.
- Let 'e_j' be eigenvectors of Q, so they form a basis

$$x^{(0)} - x^* = \sum_{j=1}^{d} \rho_j e_j$$
$$\implies Q(x^{(0)} - x^*) = \sum \lambda_j \rho_j e_j$$
$$\implies \frac{1}{2}(x^{(0)} - x^*)^T Q(x^{(0)} - x^*) = \frac{1}{2}\sum_{j=1}^{d} \lambda_j \rho_j^2$$

- Let $f(Q)$ be a matrix polynomial

$$\implies \frac{1}{2}(x^{(0)} - x^*)^T f(Q)(x^{(0)} - x^*) = \frac{1}{2}\sum_{j=1}^{d} f((\lambda_j)\rho_j^2$$

- Since, $X^{(k+1)} = argmin_{x \in B_{k+1}} f(x)$

$$\implies E(x^{(k+1)}) \leq \frac{1}{2}\sum_{j=1}^{d} \lambda_j(1 + P_k(\lambda_j)\lambda_j)^2 \rho_j^2$$
$$\max_j (1 + P_k(\lambda_j)\lambda_j)^2 = \bar{\lambda}(P_k)$$
$$E(x^{(k+1)}) \leq \frac{\bar{\lambda}}{2}\sum_{j=1}^{d} \lambda_j \rho_j^2 = \bar{\lambda}(P_k)E(x^{(0)})$$

- Suppose Q has $r < d$ distinct eigenvalues. Let the values be $\mu_1, ..., \mu_r$ $(\lambda_j = \mu_j)$
- **Strategy**: Find a polynomial $f(\lambda)$ with roots $\mu_1, ..., \mu_r$ so that $f(0) = 1$

$$f(\lambda) = \prod_{i=1}^{r} (\frac{\mu_i - \lambda}{\mu_i})$$
$$\implies f(\lambda) = 1 + P_{r-1}(\lambda)\lambda$$

- Choose $P_{r-1}(\lambda)$ so that the above equation satisfies.

$$\max_{1 \leq i \leq d} f(\lambda_i) = 0 \implies \bar{\lambda}(P_{r-1}) = 0$$
$$E(x^{(r)}) \leq 0$$

- $\implies x^{(r)}$ is the global minimum.

2.11 Quasi Newton Methods

- $f(y) = f(x) + \nabla f(x)^T(y-x) + \frac{1}{2}(y-x)^T H(x)(y-x) + o(\|y-x\|^2) = P_2 + o(\|y-x\|^2)$
$$\implies \nabla f(y) = \nabla f(x) + H(x)(y-x)$$
- We know for newton method, $u^{(k)} = -(H(x^{(k)}))^{-1} g_k$, which could be very expensive
- **Idea**: Instead of finding H_k^{-1}, find its substitute $B^{(k)}$ such that
$$\begin{aligned}
&u^{(k)} = -B^{(k)} g_k \leftarrow \text{ descent direction}\\
&x^{(k+1)} = x^{(k)} + \alpha_k u^{(k)}\\
&B^{(k+)} \to \text{update } B^{(k)} \text{ so that } B^{(k+1)} > 0 \text{ and}\\
&B^{(k+1)} r_k = \delta_k \leftarrow \text{ Quasi-Newton condition}\\
&\delta_k = -\alpha_k B^{(k)} g_k \leftarrow \text{ update rule/schema}
\end{aligned}$$
where $r_k = g_{k+1} - g_k$ and $\delta_k = x^{(k+1)} - x^{(k)}$
- **Rank-One Update**
$$\begin{aligned}
&B^{(k+1)} = B^{(k)} + tww^T\\
\implies &B^{(k+1)} r_k = B^{(k)} r_k + tww^T r_k\\
\implies &\delta_k = B^{(k)} r_k + tww^T r_k \leftarrow \text{ Quasi-Newton Condition}\\
\implies &tww^T r_k = \delta_k - B^{(k)} r_k
\end{aligned}$$
To find 'w' consider one possible solution
$$\begin{aligned}
&w = \delta_k - B^{(k)} r_k \text{ and } tw^T r_k = 1\\
\implies &t = \frac{1}{(\delta_k - B^{(k)} r_k)^T r_k}\\
&B^{(k+1)} = B^{(k)} + \frac{(\delta_k - B^{(k)} r_k)(\delta_k - B^{(k)} r_k)^T}{(\delta_k - B^{(k)} r_k)^T r_k}
\end{aligned}$$
If $\delta_0, ..., \delta_{d-1}$ are well-defined and linearly independent then the
$$B^{(d)} = Q^{-1}$$
Claim: $B^{(k+1)} r_i = \delta_i$ for $0 \le i \le k$
- **Rank-Two Update or DFP**
$$\begin{aligned}
&B^{(k+1)} = B^{(k)} + \alpha uu^T + \beta vv^T\\
&B^{(k+1)} r_k = B^{(k)} r_k + \alpha uu^T r_k + \beta vv^T r_k\\
&\delta_k = B^{(k)} r_k + \alpha uu^T r_k + \beta vv^T r_k\\
&\delta_k - B^{(k)} r_k = \alpha uu^T r_k + \beta vv^T r_k
\end{aligned}$$
One possible solution for u, v and α, β is
$$\begin{aligned}
&\alpha u^T r_k = 1,\ u = \delta_k\\
&\beta v^T r_k = -1,\ v = B^{(k)} r_k\\
&\alpha = \frac{1}{\delta_k^T r_k},\ \beta = \frac{-1}{r_k^T B^{(k)} r_k}\\
&B^{(k+1)} = B^{(k)} + \frac{\delta_k \delta_k^T}{\delta_k^T r_k} - \frac{B^{(k)} r_k r_k^T B^{(k)T}}{r_k^T B^{(k)} r_k}
\end{aligned}$$
Now, if $B^{(k)} > 0$ then $B^{(k+1)} > 0$
For our function, we can prove
$$\begin{aligned}
&\delta_i^T Q \delta_j = 0,\ 0 \le i < j \le k\\
&B^{(k+1)} Q \delta_i = \delta_i,\ 0 \le i \le k\\
\implies &B^{(k+1)} r_i = \delta_i
\end{aligned}$$
Also note $x^{(k+1)} = \delta_k + \delta_{k-1} + ... + \delta_0 + x^{(0)}$

2.12 Partial Conjugate Gradient

- Instead of using conjugate gradients to go from $x_0 \to x^*$, use it to go from $x_k \to x_{k+1}$, i.e., at each step use conjugate gradient to proceed.
- The process is carried out for $m+1 < n$ steps and then, rather than continuing, the process is restarted from the current point and $m+1$ more CG steps are taken.
- Special case of $m = 0$ corresponds to standard steepest descent.
- $min\ \frac{1}{2}x^TQx - b^Tx$
$$x_{k+1} = x_k + P_k(Q)g_k,\ g_k = Qx_k - b$$
$P_k(Q)$ is a polynomial of degree m, whose coefficients are chosen to minimize
$$E(x_{k+1}) = \frac{1}{2}(x_{k+1} - x^*)^TQ(x_{k+1} - x^*)$$
- Suppose Q has special eigenvalue structure
  1. m large eigenvalues that may or may not be located near each other
  2. $n - m$ smaller eigenvalues located within an interval $[a, b]$

  Consider the problem $min\ \frac{1}{2}x^TQx - b^Tx$ subject to $c^Tx = 0$, this can be clubbed as
$$min\ \frac{1}{2}x^TQx - b^Tx + \frac{1}{2}\mu(c^Tx)^2$$
$$\text{total quadratic term } = \frac{1}{2}x^T(Q + \mu cc^T)x$$
  thus consider eigenvalues of the matrix $Q + \mu cc^T$
  Fact: as $\mu \to \infty$, one eigenvalue $\to \infty$ and rest $n-1$ eigenvalues of matrix are in $[a, A]$
- **Theorem**: Suppose the symmetric PD matrix Q has $n-m$ eigenvalues in interval $[a, b], a > 0$ and remaining m eigenvalues are greater than b. Then the method of partial conjugate gradients, restarted every $m+1$ steps satisfies
$$E(x_{k+1}) \leq (\frac{b-a}{b+a})^2E(x_k)$$
  The point x_{k+1} is found from x_k by taking $m+1$ conjugate gradient steps so that each increment in k is a composition of several simple steps.
- **Proof**: We know from earlier
$$E(x_{k+1}) \leq \max_{\lambda_i}[1 + \lambda_i P(\lambda_i)]E(x_k)$$
  Key: Select P so that the $(m+1)$th degree polynomial $q(\lambda) = 1 + \lambda P(\lambda)$ vanishes at $\frac{a+b}{2}$ and at the m large eigenvalues of Q.
  Since $q(\lambda)$ has $m+1$ real roots
  1. $q'(\lambda)$ will have m real roots which alternate between roots of $q(\lambda)$ on real axis.
  2. $q"(\lambda)$ will have $m-1$ real roots which alternate between roots of $q'(\lambda)$

  Thus, since $q(\lambda)$ has no roots in $(-\infty, \frac{a+b}{2})$, we see that $q"(\lambda)$ doesn't change sign in that interval, also $q"(\lambda) > 0 \implies q(\lambda)$ is convex for $\lambda < \frac{a+b}{2}$.
  $\implies$ on $[0, \frac{a+b}{2}]$, $q(\lambda)$ lies below the line $1 - \frac{2\lambda}{a+b}$
$$q(\lambda) \leq 1 - \frac{2\lambda}{a+b},\ \text{on } [0, \frac{a+b}{2}]$$
$$q(\lambda) \geq 1 - \frac{2\lambda}{a+b},\ \text{on } [\frac{a+b}{2}, b]$$
$$\implies |1 + \lambda P(\lambda)| \leq |1 - \frac{2\lambda}{a+b}|,\ \text{on } [a, b]$$

3 Constrained Optimization

3.1 Introduction

- We are trying to $\min_{x \in C} f(x)$ where $C \subseteq \mathbb{R}^d$
- **Global minimum**: $\exists x^* \in C$
$$f(x^*) \leq f(X) \ \forall x \in C$$
- **Local minimum**: $\exists x^* \in C$ such that $f(x^*) \leq f(x)$
$$\forall x \in N_\delta(x^*) \cap C = \{x | \|x - x^*\| < \delta, x \in C\}$$
- All directions can't be considered in constraint optimization.
- **Feasible Direction**: At a given point $x \in C$ a vector $u \in \mathbb{R}^d$ is said to be a feasible direction, if $\exists \bar{\alpha} > 0$ such that
$$x + \alpha u \in C \ \forall 0 \leq \alpha \leq \bar{\alpha}$$
- The set of feasible directions, $FD(x) = \{u | u \text{ is a feasible direction at } x\}$
- When the point is an interior point, $FD(x)$ is similar to unconstrained case.
- If the point is on the boundary, then there are issues.
- If x^* is a local minimum then
$$\nabla f(x^*)^T u \geq 0, \ \forall u \in FD(x^*) \rightarrow \textbf{ Necessary condition}$$
- Let C be a convex se, and $x, y \in C$ then we know that
$$\alpha x + (1 - \alpha) y \in C$$
All points on the line joining x and y will be in C
- If C is a convex set, for any $y \in C$ $u = y - x$ is a feasible direction
- If 'f' is also a convex function, then at global optimum x^*,
$$\nabla f(x^*)^T (y - x^*) \geq 0, \ y \in C$$
- Consider minimizing the function $f(x) = \frac{1}{2}\|x - z\|^2$ where z is a fixed vector.
$$\nabla f(x) = x - z$$
When we optimize the given objective function, we get the closest point to z when $z \in C$, get the distance between x and z when $z \in C$.
Projection of z on to C is the closest point to z in C, this is called the **projection problem**.
- Now consider C is convex
$$f(y) = f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}\|y - x\|^2$$
$P_c(z)$: Projection of z on to C
$$P_c(z) = \arg\min_{x \in C} \frac{1}{2}\|x - z\|^2$$
$P_c(z)$ will be unique
Considering $P_c(z)$ as global optimum, then
$$f(x) = \frac{1}{2}\|x - z\|^2 \ \nabla f(x) = (x - z)$$
Now at optimum point $\nabla f(x^*) = (x^* - z) = (P_c(z) - z)$
$$\begin{aligned}
&\nabla f(x^*)^T (x - x^*) \geq 0 \\
&(P_c(z) - z)^T (x - P_c(z)) \geq 0 \\
&(P_c(z) - z)^T x \geq (P_c(z) - z)^T P_c(z) \\
&(P_c(z) - z)^T x = (P_c(z) - z)^T (P_c(z) - z) + z^T (P_c(z) - z)
\end{aligned}$$

Consider a hyperplane $a^T x = b$, it will have 2 half spaces

$$S_1 = \{x : a^T x > b\}$$
$$S_2 = \{x : a^T x < b\}$$

Take $(P_c(z) - z) = a$, we get

$$a^T x \geq \|P_c(z) - z\|^2 + a^T z$$
$$a^T x \geq a^T z, \ \forall x \in C \rightarrow \text{ Strict Inequality}$$

$\exists$ a hyperplane which separates the set C from the point z.
This is called separating **Hyper Plane Theorem**.

- Consider the function $f_0(x)$ with constraints

$$f_i(x) \geq 0, \ i = 1, ..., m$$
$$h_(x) = 0, \ i = 1, ..., n$$
$$C = \{x | f_i(x) \geq 0, \ h_j(x) = 0\}$$

Using the 1st order expansion

$$\min_{x \in \mathbb{R}^d} f_0(x_0) + \nabla f(x_0)^T (x - x_0)$$
$$f_i(x) = f_i(x_0) + \nabla f_i(x_0)^T (x - x_0)$$
$$h_j(x) = h_j(x_0) + \nabla h_j(x_0)^T (x - x_0)$$

When $f_i(x_0) > 0$

$$\nabla f_i(x_0)^T (x - x) < 0 \ \checkmark$$
$$\nabla f_i(x_0)^T (x - x) > 0 \ \checkmark$$
$$\nabla f_i(x_0)^T (x - x) = 0 \ \checkmark$$

When $f_i(x_0) = 0$

$$\nabla f_i(x_0)^T (x - x) < 0 \ \checkmark$$
$$\nabla f_i(x_0)^T (x - x) > 0 \ \checkmark$$
$$\nabla f_i(x_0)^T (x - x) = 0 \ \times$$

- Consider $A(x) = \{i | f_i(x) = 0, \ i = 1, ..., m\}$, set of active constraints at $x \in C$.
  This will generate restrictions on feasible directions.
- Feasible directions are decided more or less by active constraints.
- Feasible Descent Directions are defined as, $FDD(x^*) = \{u | \nabla f(x^*)^T u < 0, \ u \in FD(x^*)\}$
- If $FDD(x^*)$ is empty then x^* is a local minimum.

3.2 Farka's Lemma

- Consider the problem of minimizing the function $f(x)$ with constraints $g_i(x) \geq 0$
- Let $u_1, ..., u_m \in \mathbb{R}^d$ and $v \in \mathbb{R}^d$ be vectors.

$$S = \{w | w \in \mathbb{R}^d, \ w^T v < 0, w^T u_i \geq 0, \ i = 1, ..., m\}$$

This is because of constraints

$$v \approx \nabla f(x)$$
$$u_i \approx \nabla g_i(x)$$
$$w \approx \text{ vectors in } FDD$$
$$S \approx FDD(x^*)$$

S is empty if $\exists \lambda_1, ..., \lambda_m \geq 0$ such that $v = \sum_{i=1}^m \lambda_i u_i$

$$w^T u_i \geq 0, \ \forall i \in 1, ..., m$$
$$w^T v = \sum_{i=1}^m \lambda_i w^T u_i \geq 0 \implies S \text{ is empty}$$

S is empty only if $\exists \lambda_1, ..., \lambda_m \geq 0$ such that $v = \sum_{i=1}^M \lambda_i u_i$

3.3 Karush–Kuhn–Tucker conditions

- Consider the problem of minimizing the function $f(x)$ with constraints $g_i(x) \geq 0$
- Consider a function $L(x, \lambda) = f(x) - \sum_{i=1}^{m} \lambda_i g_i(x)$ where $\lambda \in \mathbb{R}^m$

$$\nabla_x L(x, \lambda) = \nabla f(x) - \sum_{i=1}^{m} \lambda_i \nabla g_i(x)$$

1. $\nabla_x L(x, \lambda) = 0$
2. $\lambda_i g_i(x) = 0$ for $i = 1, ..., m$

- If 1 and 2 are imposed together we get

$$\nabla f(x) = \sum_{i=1}^{m} \lambda_i \nabla g_i(x) = \sum_{i \in A(x)} \lambda_i \nabla g_i(x)$$

3. $g_i(x) \geq 0$, for $i = 1, ..., m \implies$ feasibility.
   As we are interested only in those points where x is feasible.
4. $\lambda_i \geq 0$ for $i = 1, ..., m$, comes from Farka's lemma

- These set of 4 conditions are called **KKT conditions**.
- If x^* is a local minimum then $\exists\ \lambda^*$ such that (x^*, λ^*) is a KKT point.
- Any pair of (x, λ) which satisfies KKT conditions are called KKT points.
- Now, problem behaves as if there are no constraints on it, minimizing is similar to unconstrained problem.
- Now, let's consider equality constraints, minimize the function $f(x)$ subject to

$$g_i(x) \geq 0,\ i = 1, ..., m$$
$$h_j(x) = 0,\ j = 1, ..., n$$

  To take care of equality constraints, we simply add 2 more inequality constraints

$$h_j(x) \geq 0 \text{ and } -h_j(x) \geq 0$$

  and our new function becomes

$$L(x, \lambda, \mu) = f(x) - \sum_{i=1}^{m} \lambda_i g_i(x) - \sum_{j=1}^{n} \mu_j h_j(x)$$

  (x^*, λ^*, μ^*) is a KKT point if

  1. $\nabla_x L(x^*, \lambda^*, \mu^*) = 0$
  2. $g_i(x^*) \geq 0$ and $h_j(x^*) = 0$
  3. $\lambda_i^* \geq 0$
  4. $\lambda_i g_i(x^*) = 0$

  No restriction on μ to be non-negative, it can be negative also.

- Consider linearized feasible direction

$$LFD(x) = \{u | \nabla g_i(x)^T u \geq 0,\ g_i(x) = 0\}$$

- The gradients of active constraints are linearly independent

$$LFD(x) = FD(x): \text{ Regular case condition}$$
$$LFD(x) \supseteq FD(x): \text{ General case}$$

- A point x is regular if $LFD(x) = FD(x)$
- KKT condition is more elaborate way of saying $\nabla f(x^*) = 0$, analogous to unconstrained case.
- Sometimes equality constraints are easy to solve by converting to unconstrained problem and use any of the previously learned methods.

3.4 Quadratic Convex Function

- Consider minimizing the function $\frac{1}{2}x^TQx + h^Tx + c$ where $Q \geq 0$ subject to

$$a_j^T x = b_j,\ j = 1, ..., n$$

We have the Lagrange equation as

$$L(x, \mu) = \frac{1}{2}x^TQx + h^Tx + c - \sum_{j=1}^{n} \mu_j(a_j^T x - b_j)$$

We have the KKT condition

$$\begin{aligned}
&\nabla_x L(x, \mu) = 0 \\
&Qx + h - \sum_{j=1}^{n} \mu_j a_j = 0 \\
&a_j^T x = b_j,\ j = 1, ..., n \\
&A = \begin{bmatrix} a_1^T \\ ... \\ a_n^T \end{bmatrix},\ b = \begin{bmatrix} b_1 \\ ... \\ b_n \end{bmatrix} \\
&\begin{bmatrix} Q & -A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x \\ \mu \end{bmatrix} = \begin{bmatrix} -h \\ b \end{bmatrix} \to 1
\end{aligned}$$

If (x^*, μ^*) solves for 1 then it is a KKT point, it is also global minimum.
Convex quadratic function subject to linear equality constraints are easy to solve.

- Consider convex quadratic function subject to linear inequality constraint(QPI)

$$\begin{aligned}
&\min_{x \in \mathbb{R}^d} \frac{1}{2}x^TQx + h^Tx + c \\
&a_j^T x \geq b_j,\ j = 1, ..., m
\end{aligned}$$

KKT conditions are

$$\begin{aligned}
&L(x, \lambda) = \frac{1}{2}x^TQx + h^Tx + c - \sum_{j=1}^{m} \lambda_j(a_j^T x - b_j) \\
&Qx + h = \sum_{j=1}^{m} \lambda_j a_j \\
&\lambda_j(a_j^T x - b_j) = 0 \\
&\lambda_j \geq 0 \\
&a_j^T x \geq b_j
\end{aligned}$$

Suppose we know $A(x^*)$ but not x^*

- If $\tilde{x}$ is solution and minimum is unique

$$\begin{aligned}
&\min_{x \in \mathbb{R}^d} \frac{1}{2}x^TQx + h^Tx + c \\
&a_j^T x = b_j\ j \in A(x^*) \\
&a_j^T \tilde{x} \geq b_j\ j \notin A(x^*) \\
&\text{then } \tilde{x} = x^*
\end{aligned}$$

- Guess $A(x^*)$, w_k guess of $A(x^*)$

$$\begin{aligned}
&w_k \subset \{1, ..., m\} \\
&f(x) = \frac{1}{2}x^TQx + b^Tx + c
\end{aligned}$$

- We have a somewhat iterative scheme as follows

  1. If $x^{(k)}$ minimizes $f(x)$ over w_k then check if $x^{(k)}$ is feasible QPI

2. If not then add infeasible constraints to w_k, minimize f, to find new $x^{(k)}$ till the $x^{(k)}$ is feasible.
3. Check the Lagrange multipliers to see if any of the Lagrange multipliers do not have proper sign. Drop constraints to find w_{k+1} and go to step 1.

- $[m] = \{1, ..., m\}$, $w_k \subseteq [m]$

$$\tilde{x}(k) \text{ is feasible } \implies a_j^T \tilde{x}^{(k)} = b_j, \ \forall j \in w_k$$
$$a_j^T \tilde{x}^{(k)} > b_j, \ j \notin w_k$$

at k^{th} iteration we know w_k is feasible, $\tilde{x}^{(k)}$ to QPI

$$f(x) = f(\tilde{x}^{(k)}) + (x - \tilde{x}^{(k)})^T (Q\tilde{x}^{(k)} + h) + \frac{1}{2}(x - \tilde{x}^{(k)})^T Q(x - \tilde{x}^{(k)})$$

Let $g_k = Q\tilde{x}^{(k)} + h$ be gradient and $x - \tilde{x}^{(k)} = u$

$$f(x) = f(\tilde{x}^{(k)}) + g_k^T u + \frac{1}{2} u^T Q u = q(u)$$

minimizing $\tilde{x}^{(k)}$ over w_k is same as minimizing u

- Let $\tilde{u} = \arg\min_u f(\tilde{x}^{(k)}) + g_k^T u + \frac{1}{2} u^T Q u$

$$a_j^T (\tilde{x}^{(k)} + u) = b_j \ j \in w_k$$
$$a_j^T u = 0$$

This is the transformed problem

- If $\tilde{u} = 0$ then $x^{(k)} = \tilde{x}^{(k)}$

$$\min_u q(u) (= \frac{1}{2} u^T Q u + g_k^T u + f(\tilde{x}^{(k)}))$$
$$a_j^T u = 0$$

Can be solved, simple equality constraint problem

- Suppose this is not true then consider $\tilde{u}$ as a feasible direction

$$\bar{x} = \tilde{x}^{(k)} + \alpha \tilde{u} \ \alpha \geq 0$$
$$a_j^T \bar{x} = a_j^T \tilde{x}^{(k)} + \alpha a_j^T \tilde{u}$$
$$= a_j^T \tilde{x}^{(k)}$$
$$= b_j$$

$j \notin w_k$

$$a_j^T \bar{x} = a_j^T \tilde{x}^{(k)} + \alpha a_j^T \tilde{u} \geq b_j$$
$$\alpha a_j^T \tilde{u} \geq b_j - a_j^T \tilde{x}^{(k)}$$

if $a_j^T \tilde{u} \geq 0$ then this inequality always holds, else

$$\alpha(-a_j^T \tilde{u}) \leq a_j^T \tilde{x}^{(k)} - b_j$$
$$\alpha \leq \frac{a_j^T \tilde{x}^{(k)} - b_j}{(-a_j^T \tilde{u})}$$
$$\bar{\alpha} = \min_{j: j \notin w_k} \frac{a_j^T \tilde{x}^{(k)} - b_j}{(-a_j^T \tilde{u})}$$

Range of α: $0 \leq \alpha \leq \bar{\alpha}$

$$\bar{x} = \tilde{x}^{(k)} + \alpha \tilde{u} \text{ is feasible}$$

if $\bar{\alpha} \geq 1$ then $x^{(k)} = \tilde{x}^{(k)} + \tilde{u}$
if $\bar{\alpha} < 1$ then $\exists j^*$ such that

$$\bar{\alpha} = \frac{a_j^T \tilde{x}^{(k)} - b_{j^*}}{(-a_{j^*}^T \tilde{u})}$$

This is a blocking constraint, $a_{j^*}^T (\tilde{x}^{(k)} + \tilde{u}) < b_{j^*}$

$$\text{update } w_k \to \{j^*\} \cup w_k$$
$$\text{update } \tilde{x}^{(k)} \to \tilde{x}^{(k)} + \bar{\alpha}\tilde{u}$$

When $\tilde{u}$ is zero this iteration will end

$$x^{(k)} = \tilde{x}^{(k)}$$

3.5 Active Set Method

- If we know w_k is a proper active set, then we can compute correct $x^{(k)}$. How to recognize it??
- Consider Quadratic Equality problem(**QP**)

$$\begin{aligned}
&\min_{x\in\mathbb{R}^d} f(x^{(k)}) + g_k^T u + \frac{1}{2}u^T Q u \\
&\quad a_j^T u = 0 \ j \in w_k \\
&\quad L(u,\mu) = \frac{1}{2}u^T Q u + g_k^T u + f(x^{(k)}) - \sum_{j\in w_k} \mu_j (a_j^T u) \\
&\quad \nabla L(0,\mu) = 0 \\
&\quad g_k = \sum_{j\in w_k} \tilde{\mu}_j a_j \\
&\quad g_k = Qx^{(k)} + h
\end{aligned}$$

$(0,\tilde{\mu})$ is a KKT point of QP.
From this can we get a candidate for original problem.
Define

$$\tilde{\lambda}_j = \begin{cases} \tilde{\mu}_j \text{ when } j \in w_k \\ 0 \text{ otherwise} \end{cases}$$

then $(x^{(k)}, \tilde{\lambda})$ a KKT point of QPI, it should satisfy KKT conditions

1. $Qx^{(k)} + h = \sum_{j=1}^m \tilde{\lambda}_j a_j$
2. $\tilde{\lambda}_j (a_j^T x^{(k)} - b_j) = 0$
3. $\tilde{\lambda}_j \geq 0$
4. $a_j^T x^{(k)} \geq b_j$

- Suppose $\exists\ c \in w_k$ such that $\tilde{\mu_c} < 0$ then $w_{k+1} = w_k$
  $\{c\}$ then resolve over w_{k+1}

$$\begin{aligned}
&\min_{\mathbb{R}^d} f(x^{(k)}) + g_k^T u + \frac{1}{2}u^T Q u \to \bar{QP} \\
&\quad a_j^T u = 0 \ j \in w_{k+1} \\
&\quad g_k = Qx^{(k)} + h \\
&\quad\quad = \sum_{j\in w_k} \tilde{\mu}_j a_j \\
&\quad\quad = \sum_{j\in w_{k+1}} \tilde{\mu}_j a_j + \tilde{\mu}_c a_c \ \tilde{\mu}_c < 0
\end{aligned}$$

Let $(\bar{u},\bar{\mu})$ be KKT point of $(\bar{QP})$

- What we did so far is removal of a constraint and evaluating the problem. This is supposed to give a better minimum.
- $u = 0$ is a feasible solution of $\bar{QP}$, we are supposed to get a better value than what is obtained at $u = 0$

$$f(x^{(k)}) + [g_k^T u + \frac{1}{2}u^T Q u] \leftarrow < 0$$

- We can say that $\bar{u}$ is a descent direction at $x^{(k)}$
- Let f_{QP} be the optimal value of QP and $f_{\bar{QP}}$ be the optimal value of $\bar{QP}$.
  Let x be feasible for QP then it is also feasible for $\bar{QP}$

$$\begin{aligned}
&f_{\bar{QP}} \leq f_{QP} \\
&\tilde{x}^{(k+1)} = x^{(k)} + \alpha\bar{u} \\
&f(\tilde{x}^{(k+1)}) \leq f(x^{(k)})
\end{aligned}$$

At $x^{(k)}$ check for global optimality, if not, identify constraints which needs to be dropped from working set.

- We can't start with null set as working set since it doesn't have feasible direction. We always need to start with at least one point that is feasible. This feasible point may not be the minimum.
- We can write the method as
  1. Find $x^{(1)}$ most likely not feasible.
  2. $w_1 = \phi$, $\tilde{x}^{(1)}$, find a feasible point
  3. Minimize f over w_1: Solve unconstrained problem
  4. Some x, in direction of u
  5. Walk along direction of u
  6. We will have a blocking constraint
  7. Add blocking constraint to working set
  8. Continue the loop till no blocking constraint $\implies \tilde{u} = 0$
  9. With λ to μ mapping, check if it is a KKT point for original problem.
  10. If no, identify which μ are negative and drop them
  11. Obtain feasible descent direction
  12. Constrict new point
  13. Repeat
- In the loop while adding and dropping constraints we may get the same working set, then **cycling** has occurred.
- In case of strict inequality
$$f(\tilde{x}^{(k+1)}) < f(x^{(k)})$$
  Cycling will not occur, when equality is added then it might.
- Solving equality constraint is the capacity required to solve this inequality case.

3.6 Gradient Projection

- Recall the following problem
$$P_c(z) = \underset{x \in C \subseteq \mathbb{R}^d}{\arg\min} \frac{1}{2}\|x - z\|^2$$
  Let C be a convex set
- Recall $\min_{x \in C} f(x) : P$, now let it be a convex function
$$\begin{aligned} &f(x) \geq f(x^*) + \nabla f(x^*)^T (x - x^*) \\ &f(x) \geq f(x^*) \end{aligned}$$
- In a strict convex function, there can't be multiple global minimum, it has to be unique.
- If Hessian is positive definite then f is strictly convex which means projection is unique.
- Coming back to projection function $P_c(z)$
$$\begin{aligned} &\nabla f(x) = x - z \\ &(P_c(z) - z)^T (x - P_c(x)) \geq 0 \to \nabla f(x)^T u \geq 0 \end{aligned}$$
  Consider two points $z^{(1)}, z^{(2)} \in \mathbb{R}^d$
$$(P_c(z^{(1)} - P_c(z^{(2)}))^T (z^{(1)} - z^{(2)}) \geq \|P_c(z^{(1)}) - P_c(z^{(2)})\|^2$$
  By Cauchy Schwarz Inequality
$$\|P_c(z^{(2)}) - P_c(z^{(1)})\| \leq \|z^{(2)} - z^{(1)}\|$$
  In the projected space, new points are closer than in the original space.
  This is called as **Non expansiveness property** of projection.

- Consider $\min_{x\in\mathbb{R}^d} \frac{1}{2}\|x-z\|^2$ subject to $Ax=b$

$$C=\{x|Ax=b\},\ C:\ \text{solution to } Ax=b$$

We consider three cases

1. $C=\phi$, then project doesn't make any sense
2. C has one element, A^{-1} exists, then also projection doesn't make any sense
3. It is interesting when C has many elements

- Consider the Lagrange equation

$$\begin{aligned}
L(x,\mu) &= \frac{1}{2}\|x-z\|^2 - \mu^T(Ax-b)\\
\nabla_x L(x,\mu) &= 0 \text{ and } Ax=b\\
x-z-\mu^T A &= 0\\
x &= z+A^T\mu\\
Ax &= Az+(AA^T\mu)=b\\
\mu &= (AA^T)^{-1}(b-Az)\\
P_c(z) &= z+A^T\mu\\
&= z+A^T(AA^T)^{-1}(b-Az)\\
&= (I-A^T(AA^T)^{-1}A)z+A^T(AA^T)^{-1}b\\
&= Qz+A^T(AA^T)^{-1}b
\end{aligned}$$

- We have the following iteration scheme

1. Start with $x^{(0)}\in C$
2. $x^{(k+1)}=P_c(x^{(k)}-\alpha_k\nabla f(x^{(k)}))$
3. $g_k=\nabla f(x^{(k)})$

- Workout P_c, similar to gradient descent
- This generates points always in C, i.e., it's a projection of gradient descent
- The constraint in the method is computing the $P_c(z)$ for which we need to compute $(AA^T)^{-1}$ once.

$$\begin{aligned}
x^{(k+1)} &= (I-A^T(AA^T)^{-1}A)(x^{(k)}-\alpha_k g_k)+A^T(AA^T)^{-1}b\\
&= x^{(k)}-A^T(AA^T)^{-1}Ax^{(k)}-\alpha_k(I-A^T(AA^T)^{-1}A)g_k+A^T(AA^T)^{-1}b\\
&= x^{(k)}-A^T(AA^T)^{-1}b-\alpha_k(I-A^T(AA^T)^{-1}A)g_k+A^T(AA^T)^{-1}b\\
&= x^{(k)}-\alpha_k(I-A^T(AA^T)^{-1}A)g_k
\end{aligned}$$

$$x^{(k+1)}=x^{(k)}-\alpha_k Q g_k$$

α_k: Step set selection can be done

- We can find step size and prove monotonic decrease using

$$f(x^{(k+1)})\le f(x^{(k)})+g_k^T(x^{(k+1)}-x^{(k)})+\frac{L}{2}\|x^{(k+1)}-x^{(k)}\|^2$$

We will get that if

$$\alpha<\frac{2}{L}\implies f(x^{(k+1)})<f(x^{(k)})$$

Algorithm will converge in N iterations

$$\begin{aligned}
f(x^{(N)})-f(x^*) &\le \frac{1}{2\alpha N}\|x^{(0)}-x^*\|^2\\
f(x^{(N)})-f(x^*) &\le \frac{L}{2\alpha N}\|x^{(0)}-x^*\|^2
\end{aligned}$$

- Apply gradient projection when projection is easy to compute or when f is convex.
- Projection may not be easy to compute always.

3.7 Linear Programming Problems

- We define the primal problem as

$$\begin{aligned} f^* = &\min_x f(x) \\ &g_i(x) \geq 0,\ i = 1, ..., m \\ &h_j(x) = 0,\ j = 1, ..., n \\ L(x, \lambda, \mu) = &f(x) - \sum_{i=1}^{m} \lambda_i g_i(x) - \sum_{j=1}^{n} \mu_j h_j(x) \end{aligned}$$

- Define $g(\lambda, \mu) = \min_x L(x, \lambda, \mu)$, is called dual function and λ, μ are dual variables.
- The value of $g(\lambda, \mu)$ can be arbitrary, can take values $-\infty$ to ∞.
- For any feasible x, i.e., $g_i(x) \geq 0, h_j(x) = 0\ \forall i, j,\ \lambda_j \geq 0$

$$\begin{aligned} &\implies L(x, \lambda, \mu) \leq f(x) \\ &\implies g(\lambda, \mu) \leq f(x)\ \forall \text{ feasible } x \text{ and } \lambda \geq 0 \\ &\implies g(\lambda, \mu) \leq f^* \end{aligned}$$

  $g(\lambda, \mu)$ is unconstrained min of $L(x, \lambda, \mu)$ and f^* is constrained min.

- We are only interested in these values, $g(\lambda, \mu) > -\infty$ to be meaningful.

$$g(\lambda, \mu) : \{(\lambda, \mu) | g(\lambda, \mu) > -\infty\} \rightarrow \text{ Dual feasible}$$

  Consider any dual feasible (λ, μ) and feasible x then $g(\lambda, \mu) \leq f(x)$ called **weak duality**.

- Consider the (λ, μ) dual feasible then

$$\max_{\lambda, \mu} g(\lambda, \mu) \text{ such that } \lambda \geq 0$$

  Let $d^* = \max g(\lambda, \mu)$ is called the dual objective function value.

$$d^* \leq f^* \text{ another definition of weak duality}$$

- Given Primal problem and dual problem, dual problem is always easy to solve. Dual is unconstrained and quadratic.

$$\begin{aligned} \text{Weak Duality: } &d^* \leq f^* \\ \text{Strong Duality: } &d^* = f^* \end{aligned}$$

- Many convex problems satisfy strong duality.
- If $\exists$ dual feasible (λ^*, μ^*) and primal feasible x^* such that $f(x^*) = g(\lambda^*, \mu^*)$
  then $f^* = f(x^*)$ and $d^* = g(\lambda^*, \mu^*)$
  x^* is not min point, its some feasible point.
  From weak duality

$$\begin{aligned} &g(\lambda^*, \mu^*) \leq f(x)\ \forall \text{ feasible } x \\ \implies &f(x^*) \leq f(x)\ \forall \text{ feasible } x \end{aligned}$$

  Hence, x^* is global minimum and $f^* = f(x^*)$.
  Similarly,

$$\begin{aligned} &g(\lambda, \mu) \leq f(x^*)\ \forall \text{ feasible } (\lambda, \mu) \\ \implies &g(\lambda, \mu) \leq g(\lambda^*, \mu^*) \end{aligned}$$

  Hence, (λ^*, μ^*) is dual optimal, $g(\lambda^*, \mu^*) = d^*$

- Suppose strong duality holds
  Solve dual problem to get (λ^*, μ^*)
  find $x^* = \arg\min_x L(x, \lambda^*, \mu^*)$

$$f(x^*) = g(\lambda^*, \mu^*) = \min_x L(x, \lambda^*, \mu^*)$$

- Strong Duality holds for many convex problems
  Suppose Linear Problems holds this property

$$\min_{x\in\mathbb{R}^d} c^T x \quad \max_{\mu\in\mathbb{R}^m} b^T\mu$$
$$Ax = b \; A^T\mu \le c$$
$$x \ge 0$$

  Find the KKT point for primal

$$L(x,\lambda,\mu) = c^T x - \mu^T(Ax-b) - \lambda^T x$$

  Let $(x^{(0)},\lambda^{(0)},\mu^{(0)})$ be KKT point

$$g(x^{(0)},\lambda^{(0)},\mu^{(0)}) = (c - A^T\mu - \lambda)^T x^{(0)} + b^T\mu^{(0)} = b^T\mu^{(0)}$$
$$L(x^{(0)},\lambda^{(0)},\mu^{(0)}) = c^T x^{(0)} - \mu^T(Ax^{(0)} - b) - \lambda^{(0)T}x^{(0)} = c^T x^{(0)}$$
$$\implies b^T\mu^{(0)} = c^T x^{(0)}$$

  Since, $\mu^{(0)}$ is dual feasible and $x^{(0)}$ is primal
  feasible $\implies$ Strong duality holds.

- P is a constrained problem, we can solve P by solving D its dual problem which is unconstrained.
- When strong duality doesn't hold, then for a feasible $(x^{(k)},\lambda^{(k)},\mu^{(k)})$ of primal and dual

$$f(x^{(k)}) \ge g(\lambda^{(k)},\mu^{(k)})$$
$$f(x^{(k)}) - g(\lambda^{(k)},\mu^{(k)}) > 0 \to \text{ Duality Gap}$$

  find points such that we have the least duality gap.

- If a variable is only in the constraint and no impact on the objective, we can remove that from constraints and replace by inequality.
- Any number can be written as a difference of 2 positive numbers.
- Solution is unique, then it is obtained at corner.
- If the solution is not unique then answer is at corners and some other points.
- Consider the LP problem

$$\min_{x\in\mathbb{R}^d} c^T x$$
$$Ax = b \text{ and } x \ge 0$$
$$A \in \mathbb{R}^{m\times d} \; c\in\mathbb{R}^d \; b\in\mathbb{R}^m \; m < d$$

  Constraint set $C = \{z | Az = b,\ z \ge 0\}$
  $x \in C \implies x$ is feasible solution.
  Consider A is full rank $\implies$ rank$(A) = m \implies \exists m$ LI columns of A.
  Let B be the LI columns of A, and N be the rest

$$AX = \begin{bmatrix} B & N \end{bmatrix} \begin{bmatrix} X_B \\ X_N \end{bmatrix} = BX_B + NX_N = b$$

  $X_N = 0$, B is full rank $\implies$ invertible $\implies X_B = B^{-1}b$

$$\implies X = \begin{bmatrix} B^{-1}b \\ 0 \end{bmatrix} \text{ is the solution}$$

  B forms the basis of A, B is not unique.
  We say in the basis of B, X is the basic feasible solution.
  There will be d_{C_m} solution at most each pertaining to a choice of "m" vectors.

- If $X \ge 0$ then x is basic feasible solution.

- Let $\hat{x} \in C$ $\hat{x} = \begin{bmatrix} \hat{x}_1 \\ ... \\ \hat{x}_d \end{bmatrix}$
  where $\hat{x}_i > 0$ for $i = 1, ..., l$ and $= 0$ for $i = l+1, ..., d$
  Since, $\hat{x} \in C$ $A\hat{x} = b$
  Suppose $A_1, ..., A_l$ are LI $\implies l \leq m$
  If $l = m \implies \sum_{i=1}^{m} \hat{x}_i A_i = b$ forms a basis and $\hat{x}$ is BFS
  If $l < m$, $A_1, ..., A_l$ is not a basis. Then find $m - l$ columns from $A_{l+1}, ..., A_d$.
  Suppose $A_1, ..., A_l$ are not LI, then we can reduce to LI. Once, the columns are reduces, we consider case 1

- If there is a feasible solution, then there is a basic feasible solution.

- If there is a optimal feasible solution then there is an optimal BFS.

- For a convex set $C \subseteq \mathbb{R}^d$
  $x \in C$ is an extreme point of C if there are no 2 distinct points $x_1, x_2 \in C$ such that

$$(1-\alpha)x_1 + \alpha x_2 \in C$$
$$x_1 \neq x_2 \neq x$$

- x is an extreme point, then x is a BFS.

- Finding one BFS is not difficult but scanning over all BFS will be an issue.

- Proposed procedure
  Is BFS optimal, if not find new BFS.

- Guess $x^{(0)}$, $x^{(0)}$ provides B
  Compute all $c_{j+m} - c_B^T \hat{A}_{j+m} \geq 0$ conditions.
  If condition satisfies then x is optimal return it.
  If not, then $\exists L$ such that $c_L - c_B^T B^{-1} A_L < 0$
  Improve function value $x(\alpha) = x^{(0)} + \alpha e_L$ and $z = \alpha e_j$
  This is LP-R algorithm.

3.8 Simplex Method

- Initialize: Given Basis B
  Find $x_B = \hat{b}, x_N = 0$ $x = \begin{bmatrix} x_B \\ x_N \end{bmatrix}$
  Compute $c_j - c_B^T \hat{A}_j$ where $\{j : A_j \notin B\}$
  Find L, $c_L - c_B^T \hat{A}_L < 0$ and $\hat{A}_j = B^{-1} A_j$
  Find $k = \arg\min_i \frac{\hat{b}_i}{\hat{A}_{iL}}$ $\hat{A}_{iL} > 0$
  Update basis, replace A_k by A_L then go back to step 1.

- Can't find L

$$c_j - c_B^T B^{-1} A_j \geq 0 \ \forall j$$

  BFS x with basis B is optimal

- Can't find k, $\hat{A}_{ik} \leq 0$
  Original problem unbounded, $\bar{\alpha}$ is unbounded.

- Simplex algorithm converges if there is no degeneracy($\hat{b}_i \neq 0 \ \forall i$), strict decrease will happen each step.